Model Architecture: Qwen3-0.6B

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1 Model Overview

Model name	Qwen3-0.6B
Transformers class	`Qwen3ForCausalLM`
Path	`/models/Qwen3-0.6B`
Total parameters	596,049,920 (596.05M)
Weight dtype	bfloat16
Hidden size d	1,024
Decoder layers L	28
Query heads H	16
Key/value heads H_{kv}	8
Head dim d_h	128
MLP width d_{ff}	3,072
Vocabulary V	151,936
Max context length	40,960
RMSNorm ϵ	1e-06
Tied input/output embeddings	True
Activation	SiLU
RoPE base θ	1,000,000

How to read this document. Section 2 treats the model mathematically, top-down: first the whole network as one composition $\Psi \circ \mathcal{B}^{(L)} \circ \cdots \circ \mathcal{B}^{(1)} \circ E$ of layers, then a definition of every layer down to single neurons and attention heads, then parameters, outputs, intermediate states and the local linear response. Section 3 gives readable pseudocode for the forward pass and for every distinct block type. Section 4 lists the weights of each block type *once*, with how many times it repeats. Section 5 explains in plain language what every kind of layer does. The original layer-by-layer table with every repetition written out is in Appendix A.

Basic notation

B, T	batch size, sequence length (tokens)
d	hidden size (1024)
V	vocabulary size (151,936)
H, H_{kv}, d_h	query heads, key/value heads, size of one head
W_*, b_*	learned weight matrix / bias of the named projection
$\odot$, σ	element-wise product, logistic sigmoid
$f^{\circ n}$	f applied n times in a row ($f \circ f \circ \cdots \circ f$)
$g \circ f$	first f, then g (composition reads right to left)
RMSNorm(x)	$\gamma \odot x / \sqrt{\text{mean}(x^2) + \epsilon}$ (learned scale γ)
LN(x)	LayerNorm: $\gamma \odot (x - \mu)/\sigma + \beta$
RoPE	rotary position embedding (rotates query/key pairs by an angle set by position)

2 Mathematical Representation

We describe the model *top-down*. First the whole network is written as one composition of functions (Section 2.1). Then every function that appears in that composition is defined (Section 2.2), down to individual neurons and attention heads. The remaining subsections describe the parameters, the outputs, the intermediate states, and the local linear behaviour.

2.1 The model as a composition of layers

Definition 2.1 (Qwen3-0.6B as an L-layer decoder)**.** The model is a function $f_\theta : \mathcal{X} \to \mathcal{Y}$ with parameters θ. It maps a sequence of T token ids $x \in \mathcal{X} = \{1, \ldots, V\}^T$ to one logit vector per position, $y \in \mathcal{Y} = \mathbb{R}^{T \times V}$ ($V = 151{,}936$). It is a composition of an embedding E, $L = 28$ decoder blocks $\mathcal{B}^{(\ell)}$ and an output head Ψ:

$$f_\theta \;=\; \Psi \circ \mathcal{B}^{(L)} \circ \mathcal{B}^{(L-1)} \circ \cdots \circ \mathcal{B}^{(2)} \circ \mathcal{B}^{(1)} \circ E \tag{1}$$

Every block maps $\mathbb{R}^{T \times d} \to \mathbb{R}^{T \times d}$ ($d = 1024$), so the blocks can be stacked in any number. Block ℓ has its own parameters θ_ℓ and is stored in the checkpoint as `model.layers.`$(\ell - 1)$.

Definition 2.2 (Block types)**.** All L blocks have the same functional form $\mathcal{B}_{\mathrm{F}}$ (Full (global) attention block). They differ only in their weights: $\mathcal{B}^{(\ell)} = \mathcal{B}_{\mathrm{F}}(\,\cdot\,;\theta_\ell)$. Written out, with the type as a subscript:

$$f_\theta = \Psi \circ \mathcal{B}_{\mathrm{F}}^{(28)} \circ \cdots \circ \mathcal{B}_{\mathrm{F}}^{(3)} \circ \mathcal{B}_{\mathrm{F}}^{(2)} \circ \mathcal{B}_{\mathrm{F}}^{(1)} \circ E \tag{2}$$

Grouping the repetitions with composition powers ($\mathcal{B}^{\circ n} = n$ consecutive blocks of the same type, each with its own weights) gives the compact form

$$f_\theta = \Psi \circ \left(\mathcal{B}_{\mathrm{F}}^{\circ 28}\right) \circ E \tag{3}$$

Layer pattern.

Symbol	Block type	Count	Params / block	Checkpoint indices
$\mathcal{B}_{\mathrm{F}}$	Full (global) attention block	28	15,730,944	0–27

Language-model decoder (28 layers). layers 0–27: $[\mathcal{B}_{\mathrm{F}}]$ repeated 28× (checkpoint indices, i.e. $\ell - 1$).

2.2 The layers, top to bottom

Every function in the composition above is defined here. Only the attention (or linear-attention) sub-layer mixes information *between* positions t. Every other sub-layer (norms, feed-forward networks, projections) acts on each token vector $h_t \in \mathbb{R}^d$ independently, e.g. $\mathrm{FFN}(h)_t = \mathrm{FFN}(h_t)$. All weights follow the convention $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, so a projection of a token vector is Wu (as in an affine layer $f_i(u) = A^{(i)}u + b^{(i)}$).

Definition 2.3 (Token embedding)**.** $E : \{1, \ldots, V\}^T \to \mathbb{R}^{T \times d}$ looks up one row of the embedding matrix $W_E \in \mathbb{R}^{V \times d}$ per token:

$$E(x)_t = W_E[x_t, :], \qquad t = 1, \ldots, T. \tag{4}$$

Definition 2.4 (Normalisation)**.** $\mathrm{RMSNorm}(u) = \gamma \odot u / \sqrt{\frac{1}{d}\sum_i u_i^2 + \epsilon}$, with a learned scale $\gamma \in \mathbb{R}^d$, applied to each token vector separately. It fixes the scale of a vector without changing its direction. Every norm in the model has its own γ, distinguished by a subscript (e.g. $\mathrm{RMSNorm}_1$, $\mathrm{RMSNorm}_{\text{final}}$).

Definition 2.5 (Block $\mathcal{B}_{\mathrm{F}}$: Full (global) attention block)**.** A decoder block of this type maps $h \in \mathbb{R}^{T \times 1024}$ to the same space. It is a composition of 2 residual sub-layers, each of the form $(\mathrm{id} + g)(h) = h + g(h)$:

$$\mathcal{B}_{\mathrm{F}} = (\mathrm{id} + \mathrm{FFN} \circ \mathrm{RMSNorm}_2) \circ (\mathrm{id} + \mathrm{Attn} \circ \mathrm{RMSNorm}_1) \tag{5}$$

Step by step, the sub-layers are:

$$\begin{aligned}
a &= h + \mathrm{Attn}(\mathrm{RMSNorm}_1(h)) && \text{pre-norm + residual}\\
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{RMSNorm}_2(a)) && \text{pre-norm + residual}\\
Q &= W_Q\, u && H = 16 \text{ heads of size } d_h = 128\\
K,\ V &= W_K\, u, \quad W_V\, u && H_{kv} = 8 \text{ key/value heads}\\
Q,\ K &= \mathrm{RMSNorm}_q(Q),\ \mathrm{RMSNorm}_k(K) && \text{per-head normalisation}\\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}\\
S_h &= s\, Q_h K_{g(h)}^\top + M && \text{mask } M\text{: causal}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\, V_{g(h)}\\
\mathrm{Attn}(u) &= W_O\, [O_1; \ldots; O_H]\\
\mathrm{FFN}(u) &= W_{\text{down}}\big(\,\mathrm{SiLU}(W_{\text{gate}}\, u) \odot W_{\text{up}}\, u\big) && \text{gated MLP, width 3072}
\end{aligned}$$

Parameters of one block ($|\theta_\ell| = 15{,}730{,}944$): $W_Q \in \mathbb{R}^{2048\times1024}$, $W_K \in \mathbb{R}^{1024\times1024}$, $W_V \in \mathbb{R}^{1024\times1024}$, $W_O \in \mathbb{R}^{1024\times2048}$, $W_{\text{gate}} \in \mathbb{R}^{3072\times1024}$, $W_{\text{up}} \in \mathbb{R}^{3072\times1024}$, $W_{\text{down}} \in \mathbb{R}^{1024\times3072}$, plus 2,304 parameters in norm scales, biases and other vectors. This type is used 28 times.

Definition 2.6 (Output head)**.** $\Psi : \mathbb{R}^{T\times d} \to \mathbb{R}^{T\times V}$ normalises the last hidden state and scores every vocabulary token, re-using the embedding matrix (tied weights):

$$\Psi(h)_t = W_E^\top\, \mathrm{RMSNorm}_{\text{final}}(h)_t \tag{6}$$

2.3 Neurons and attention heads

Definition 2.7 (Neuron, pre-activation value and state)**.** The feed-forward sub-layer of block ℓ contains $d_{ff}^{(\ell)} = 3072$ neurons. Neuron j of layer ℓ is the scalar function of a normalised token vector $u \in \mathbb{R}^d$

$$\eta_j^{(\ell)}(u) = \mathrm{SiLU}\left(W_{\mathrm{gate},j}^{(\ell)} \cdot u\right)\left(W_{\mathrm{up},j}^{(\ell)} \cdot u\right), \tag{7}$$

where $W_{\mathrm{gate},j}$ is the j-th row of W_{gate}. The feed-forward output is the sum of the neuron outputs, each written along its own column of W_{down}:

$$\mathrm{FFN}_\ell(u) = \sum_{j=1}^{d_{ff}^{(\ell)}} \eta_j^{(\ell)}(u)\, W_{\mathrm{down}}^{(\ell)}[:, j]. \tag{8}$$

Across its L blocks the network has $N = \sum_{\ell=1}^{L} d_{ff}^{(\ell)} = 86{,}016$ feed-forward neurons. For input x at position t, with $u_t^{(\ell)}(x)$ the normalised input of the feed-forward sub-layer:

- $\mathcal{V}(\eta; x, t) = W_{\mathrm{gate},j}^{(\ell)} \cdot u_t^{(\ell)}(x) \in \mathbb{R}$ is the **pre-activation value** of neuron η (the argument of the non-linearity).
- $\mathcal{S}(\eta; x, t)$ is its **activation state**: *active* if $\mathcal{V} > 0$ and *inactive* if $\mathcal{V} \le 0$. The non-linearity is smooth, unlike ReLU, so an inactive neuron's output is strongly suppressed rather than exactly zero.

Definition 2.8 (Attention head and attention pattern)**.** Head h of an attention layer ℓ reads from the whole sequence through its attention pattern $P_h^{(\ell)}(x) \in \mathbb{R}^{T \times T}$ (row t = how much position t attends to each earlier position) and writes through its own $d \times d_h$ slice $W_{O,h}^{(\ell)}$ of W_O:

$$\mathrm{Attn}_\ell(u) = \sum_{h=1}^{H} P_h^{(\ell)}\, V_{g(h)}\, W_{O,h}^{(\ell)\top}. \tag{9}$$

$g(h)$ is the key/value head shared by query head h (grouped-query attention). The pattern P_h plays the role that the activation state plays for neurons: once P_h is fixed, the head is linear in its values. The network has 28 attention layers $\times$ $H = 16$ heads $= 448$ attention heads.

2.4 Architecture and parameters

Definition 2.9 (Architecture and parameters)**.** The **architecture** is the tuple of sizes and the layer-type sequence:

$$\mathcal{A} = (V, d, L, c, H, H_{kv}, d_h, d_{ff}) = (151{,}936,\ 1024,\ 28,\ (\mathrm{F}^{28}),\ 16,\ 8,\ 128,\ 3072). \tag{10}$$

The **parameters** are $\theta = \{W_E,\ \theta_1, \ldots, \theta_L,\ \gamma_{\mathrm{final}}\}$ (W_E is shared with the output head). Grouping the blocks by type,

$$\begin{aligned} |\theta| &= \sum_{\text{types } c} n_c\, |\theta_c| + |\theta_{\mathrm{rest}}| \\ &= 28 \times 15{,}730{,}944 + 155{,}583{,}488 \\ &= 596{,}049{,}920, \end{aligned}$$

where n_c is how often type c occurs and θ_{rest} holds everything outside the repeated stacks (embedding, final norm, …; Table 3).

2.5 Output representations

Definition 2.10 (Output representations)**.** For an input sequence x, the output can be read at three levels:

1. **Logits:** $y = f_\theta(x) \in \mathbb{R}^{T \times V}$, one score per vocabulary token at every position.
2. **Next-token distribution:** $p(x)_{t,k} = \dfrac{\exp(y_{t,k})}{\sum_{j=1}^{V} \exp(y_{t,j})} = \Pr\left(x_{t+1} = k \mid x_1, \ldots, x_t\right)$.
3. **Predicted next token:** $z(x) = \mathrm{argmax}_{k \in \{1,\ldots,V\}}\, y_{T,k}$ (greedy decoding). Generation appends $z(x)$ (or a sample from $p(x)_T$) to x and applies f_θ again; in practice the keys and values of earlier positions are cached.

The model is **causal**: y_t depends only on $x_1, \ldots, x_t$, because every attention mask hides later positions. Training minimises the cross-entropy $-\sum_t \log p(x)_{t, x_{t+1}}$.

2.6 Intermediate states and sub-network decomposition

To study what happens at one layer $\ell \in \{1, \dots, L\}$, split the composition into a prefix, the layer itself and a suffix:

$$f_\theta = \underbrace{\Psi \circ \mathcal{B}^{(L)} \circ \cdots \circ \mathcal{B}^{(\ell+1)}}_{G_{\ell+1}\ \text{(suffix)}} \circ \underbrace{\mathcal{B}^{(\ell)}}_{\text{layer } \ell} \circ \underbrace{\mathcal{B}^{(\ell-1)} \circ \cdots \circ \mathcal{B}^{(1)} \circ E}_{F_{\ell-1}\ \text{(prefix)}} \tag{11}$$

- **Prefix** $F_{\ell-1} : \mathcal{X} \to \mathbb{R}^{T \times d}$ maps the tokens to the hidden state entering layer ℓ ($F_0 = E$).
- **Hidden state** (residual stream) after layer ℓ: $h^{(\ell)} = F_\ell(x) = \mathcal{B}^{(\ell)}(h^{(\ell-1)}) \in \mathbb{R}^{T \times d}$, with $h^{(0)} = E(x)$.
- **Inside layer ℓ:** the attention write $a^{(\ell)} = \mathrm{Attn}_\ell\big(\mathrm{RMSNorm}_1(h^{(\ell-1)})\big)$, the mid-block state $\hat{h}^{(\ell)} = h^{(\ell-1)} + a^{(\ell)}$, the feed-forward input $u^{(\ell)} = \mathrm{RMSNorm}_2(\hat{h}^{(\ell)})$, the pre-activations $\hat{x}^{(\ell)} = u^{(\ell)} W_{\text{gate}}^{(\ell)\top} \in \mathbb{R}^{T \times d_{ff}}$ and the neuron outputs $\eta^{(\ell)}(u^{(\ell)}) \in \mathbb{R}^{T \times d_{ff}}$, and the feed-forward write $m^{(\ell)} = \mathrm{FFN}_\ell(u^{(\ell)})$.
- **Suffix** $G_{\ell+1} : \mathbb{R}^{T \times d} \to \mathbb{R}^{T \times V}$ maps $h^{(\ell)}$ to the logits ($G_{L+1} = \Psi$).

Residual stream. Because every sub-layer only *adds* to the hidden state, unrolling the composition gives the final hidden state as the embedding plus everything the layers wrote:

$$h^{(L)} = h^{(0)} + \sum_{\ell=1}^{L} \big(a^{(\ell)} + m^{(\ell)}\big) \tag{12}$$

So $f_\theta(x) = \Psi(h^{(L)})$: each layer reads the current stream and adds a correction to it.

2.7 Local linear representation

A ReLU network is piecewise affine: inside a polytope cell it collapses to one matrix. This model is not, because softmax, SiLU and the normalisations are smooth. Its local behaviour is described by the first-order (Jacobian) linearisation instead. Tokens are discrete, so we perturb the embedded input $h^{(0)}$.

Definition 2.11 (Local linear response)**.** Let $J_\ell = D\mathcal{B}^{(\ell)}(h^{(\ell-1)})$ be the Jacobian of block ℓ at the current hidden state (tensors flattened to vectors). The chain rule applied to the residual sub-layers $(\mathrm{id} + g_{\ell,i})$ gives

$$J_\ell = (I + Dg_{\ell,2})(I + Dg_{\ell,1}), \tag{13}$$

where $g_{\ell,1}$ is the attention branch, $g_{\ell,2}$ the feed-forward branch and each Dg is taken at that branch's input. For a small perturbation δ of $h^{(0)}$, the response of the network decomposed around layer ℓ is

$$f_\theta(h^{(0)} + \delta) - f_\theta(h^{(0)}) \approx G_x^{(\ell+1)} \cdot J_\ell \cdot F_x^{(\ell-1)} \cdot \delta, \tag{14}$$

with $F_x^{(\ell-1)} = J_{\ell-1} \cdots J_1$ (prefix) and $G_x^{(\ell+1)} = J_\Psi\, J_L \cdots J_{\ell+1}$ (suffix). Multiplying out the products of $(I + Dg)$ terms writes $J_\Psi J_L \cdots J_1$ as a sum over *paths*: the identity path straight from embedding to head, plus one term for every subset of sub-layers a signal passes through.

2.8 Summary of notation

Symbol	Meaning
$f_\theta : \mathcal{X} \to \mathcal{Y}$	the whole model, $\{1..V\}^T \to \mathbb{R}^{T \times V}$
$E,\ \Psi$	embedding and output head
$\mathcal{B}^{(\ell)},\ \theta_\ell$	block ℓ and its parameters, $\ell = 1, \dots, L$
$\mathcal{B}_c,\ c(\ell)$	block type c; type of layer ℓ
$\mathcal{B}^{\circ n}$	n consecutive blocks of one type (each with its own weights)
$(\mathrm{id} + g)$	residual sub-layer $h \mapsto h + g(h)$
$h^{(\ell)}$	hidden state (residual stream) after layer ℓ; $h^{(0)}$ = embeddings
$a^{(\ell)},\ m^{(\ell)}$	what the attention / feed-forward sub-layer of layer ℓ adds to the stream
$\eta_j^{(\ell)}$	j-th feed-forward neuron of layer ℓ
$\mathcal{V}(\eta; x, t),\ \mathcal{S}(\eta; x, t)$	pre-activation value and active/inactive state of a neuron
$P_h^{(\ell)}$	attention pattern of head h in layer ℓ ($T \times T$)

continued on next page

Symbol	Meaning
$F_{\ell-1},\ G_{\ell+1}$	prefix (embedding to layer $\ell-1$) and suffix (layer $\ell+1$ to logits)
$J_\ell,\ F_x^{(\ell-1)},\ G_x^{(\ell+1)}$	Jacobian of block ℓ, and of the prefix / suffix at input x
$p(x),\ z(x)$	next-token distribution and greedy next-token prediction

3 Forward Pass in Pseudocode

Algorithm 1 Qwen3-0.6B: full forward pass

Require: token ids x of shape $B \times T$

$h \leftarrow E(x)$ ▷ $h^{(0)}$: one vector per token, $B{\times}T{\times}1024$

for $\ell \leftarrow 1$ **to** 28 **do** ▷ 28 blocks of type $\mathcal{B}_{\mathrm{F}}$

  $h \leftarrow \mathcal{B}_{\mathrm{F}}(h;\, \theta_\ell)$ ▷ Full (global) attention block; result is $h^{(\ell)}$

end for

$h \leftarrow \mathrm{RMSNorm}_{\mathrm{final}}(h)$ ▷ $h = h^{(L)}$; this and the next step are Ψ

$y \leftarrow h\, W_E^\top$ ▷ logits $B{\times}T{\times}151,936$, embedding matrix re-used

return y ▷ softmax(y) = next-token probabilities

Algorithm 2 Block $\mathcal{B}_{\mathrm{F}}(h)$ — Full (global) attention block

Require: h of shape $B \times T \times 1024$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection

$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector

token mixing: multi-head self-attention

$Q \leftarrow uW_Q^\top$ ▷ $B{\times}T{\times}16{\times}128$ (16 query heads)

$K \leftarrow uW_K^\top;\ V \leftarrow uW_V^\top$ ▷ $B{\times}T{\times}8{\times}128$ (8 key/value heads, each shared by 2 query heads)

$Q \leftarrow \mathrm{RMSNorm}(Q);\ K \leftarrow \mathrm{RMSNorm}(K)$ ▷ normalise each head vector (stabilises training)

$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs

$P \leftarrow \mathrm{softmax}(s\, QK^\top + M)$ ▷ mask: causal: a token sees only earlier tokens

$O \leftarrow PV$ ▷ $B{\times}T{\times}16{\times}128$

$m \leftarrow O\, W_O^\top$ ▷ concatenate heads, project back to $B{\times}T{\times}1024$

$h \leftarrow r + m$ ▷ add residual

channel mixing: feed-forward network (each token independently)

$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$

$m \leftarrow \big(\mathrm{SiLU}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big) W_{\mathrm{down}}^\top$ ▷ expand to 3072, gate, shrink back to 1024

$h \leftarrow r + m$ ▷ add residual

return h

4 Compact Architecture (repeated layers written once)

4.1 Modules outside the repeated stacks

Table 3: Weights that appear once

Weight	Type	Shape	Params
`model.embed_tokens.weight`	Embedding	151936×1024	155,582,464
`model.norm.weight`	Qwen3RMSNorm	1024	1,024
`lm_head.weight` (tied to `model.embed_tokens.weight`)	Linear	151936×1024	0 (shared)

4.2 Language-model decoder: `model.layers`

Table 4: $\mathcal{B}_{\mathrm{F}}$: Full (global) attention block — layers 0–27; weight names are relative to `model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	2048×1024	2,097,152
`self_attn.k_proj.weight`	Linear	1024×1024	1,048,576
`self_attn.v_proj.weight`	Linear	1024×1024	1,048,576
`self_attn.o_proj.weight`	Linear	1024×2048	2,097,152
`self_attn.q_norm.weight`	Qwen3RMSNorm	128	128
`self_attn.k_norm.weight`	Qwen3RMSNorm	128	128
`mlp.gate_proj.weight`	Linear	3072×1024	3,145,728
`mlp.up_proj.weight`	Linear	3072×1024	3,145,728
`mlp.down_proj.weight`	Linear	1024×3072	3,145,728
`input_layernorm.weight`	Qwen3RMSNorm	1024	1,024
`post_attention_layernorm.weight`	Qwen3RMSNorm	1024	1,024
Per block			**15,730,944**
× 28 blocks			**440,466,432**

5 What Each Layer Does

One row per *kind* of layer present in this model. "Appears as" gives the module names (layer indices removed).

Layer (appears as)	What it does
Language model	
Token embedding `embed_tokens`	Lookup table with one learned vector per vocabulary token. Turns token ids into the vectors the network computes with (Gemma multiplies them by $\sqrt{d}$).
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?".
Query norm (q_norm) `self_attn.q_norm`	RMSNorm applied to every query head vector before RoPE. Keeps attention logits in a sane range (QK-norm).
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Key norm (k_norm) `self_attn.k_norm`	RMSNorm applied to every key head vector before RoPE.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.
Output projection (o_proj) `self_attn.o_proj`	Concatenates all heads' results and mixes them back into one hidden-size vector per token.
Gate projection (gate_proj) `mlp.gate_proj`	Expands the token vector to the MLP width and passes it through SiLU/GELU. The result acts as a soft on/off switch for each hidden unit.
Up projection (up_proj) `mlp.up_proj`	Expands the token vector to the MLP width (the "content" path). It is multiplied element-wise by the gate (SwiGLU/GeGLU).
Down projection (down_proj) `mlp.down_proj`	Shrinks the gated hidden vector back to the model width so it can be added to the residual stream.
Input norm (input_layernorm) `layers.input_layernorm`	RMSNorm applied to the residual stream before attention. Rescales each token vector to unit RMS, then multiplies by a learned per-channel weight.
post_attention_layernorm `layers.post_attention_layernorm`	Despite the name, this is the pre-MLP norm: RMSNorm on the residual stream after attention, feeding the MLP.
Final norm `norm`	RMSNorm applied once after the last decoder layer, just before the LM head.

6 Parameter Count

Table 6: Parameters by component

Component	Parameters	% of total
MLP / MoE	264,241,152	44.3%
Attention / token mixer	176,167,936	29.6%
Token embedding	155,582,464	26.1%
Norms	58,368	0.0%
Total	**596,049,920**	**100.0%**

Table 7: Parameters by block type

Block type	Count	Params / block	Total
$\mathcal{B}_{\mathrm{F}}$ Full (global) attention block	28	15,730,944	440,466,432
Modules outside stacks	1	–	155,583,488
Total			**596,049,920**

Appendix A Full Layer-by-Layer Table

Every weight tensor with every repetition written out (the original table).

Layer path	Type	Shape	Dtype	Params
`lm_head.weight` *(tied)*	Linear	151936 × 1024	bfloat16	155,582,464
`model`	Qwen3Model	–	–	–
`model.embed_tokens.weight`	Embedding	151936 × 1024	bfloat16	155,582,464
`model.layers.0`	Qwen3DecoderLayer	–	–	–
`model.layers.0.input_layernorm.weight`	Qwen3RMSNorm	1024	bfloat16	1,024
`model.layers.0.mlp`	Qwen3MLP	–	–	–
`model.layers.0.mlp.down_proj.weight`	Linear	1024 × 3072	bfloat16	3,145,728
`model.layers.0.mlp.gate_proj.weight`	Linear	3072 × 1024	bfloat16	3,145,728
`model.layers.0.mlp.up_proj.weight`	Linear	3072 × 1024	bfloat16	3,145,728
`model.layers.0.post_attention_layernorm.weight`	Qwen3RMSNorm	1024	bfloat16	1,024
`model.layers.0.self_attn`	Qwen3Attention	–	–	–
`model.layers.0.self_attn.k_norm.weight`	Qwen3RMSNorm	128	bfloat16	128
`model.layers.0.self_attn.k_proj.weight`	Linear	1024 × 1024	bfloat16	1,048,576
`model.layers.0.self_attn.o_proj.weight`	Linear	1024 × 2048	bfloat16	2,097,152
`model.layers.0.self_attn.q_norm.weight`	Qwen3RMSNorm	128	bfloat16	128
`model.layers.0.self_attn.q_proj.weight`	Linear	2048 × 1024	bfloat16	2,097,152
`model.layers.0.self_attn.v_proj.weight`	Linear	1024 × 1024	bfloat16	1,048,576
`model.layers.1`	Qwen3DecoderLayer	–	–	–
`model.layers.1.input_layernorm.weight`	Qwen3RMSNorm	1024	bfloat16	1,024
`model.layers.1.mlp`	Qwen3MLP	–	–	–
`model.layers.1.mlp.down_proj.weight`	Linear	1024 × 3072	bfloat16	3,145,728
`model.layers.1.mlp.gate_proj.weight`	Linear	3072 × 1024	bfloat16	3,145,728
`model.layers.1.mlp.up_proj.weight`	Linear	3072 × 1024	bfloat16	3,145,728
`model.layers.1.post_attention_layernorm.weight`	Qwen3RMSNorm	1024	bfloat16	1,024
`model.layers.1.self_attn`	Qwen3Attention	–	–	–
`model.layers.1.self_attn.k_norm.weight`	Qwen3RMSNorm	128	bfloat16	128
`model.layers.1.self_attn.k_proj.weight`	Linear	1024 × 1024	bfloat16	1,048,576
`model.layers.1.self_attn.o_proj.weight`	Linear	1024 × 2048	bfloat16	2,097,152
`model.layers.1.self_attn.q_norm.weight`	Qwen3RMSNorm	128	bfloat16	128
`model.layers.1.self_attn.q_proj.weight`	Linear	2048 × 1024	bfloat16	2,097,152
`model.layers.1.self_attn.v_proj.weight`	Linear	1024 × 1024	bfloat16	1,048,576
`model.layers.2`	Qwen3DecoderLayer	–	–	–
`model.layers.2.input_layernorm.weight`	Qwen3RMSNorm	1024	bfloat16	1,024
`model.layers.2.mlp`	Qwen3MLP	–	–	–
`model.layers.2.mlp.down_proj.weight`	Linear	1024 × 3072	bfloat16	3,145,728
`model.layers.2.mlp.gate_proj.weight`	Linear	3072 × 1024	bfloat16	3,145,728
`model.layers.2.mlp.up_proj.weight`	Linear	3072 × 1024	bfloat16	3,145,728
`model.layers.2.post_attention_layernorm.weight`	Qwen3RMSNorm	1024	bfloat16	1,024
`model.layers.2.self_attn`	Qwen3Attention	–	–	–
`model.layers.2.self_attn.k_norm.weight`	Qwen3RMSNorm	128	bfloat16	128
`model.layers.2.self_attn.k_proj.weight`	Linear	1024 × 1024	bfloat16	1,048,576
`model.layers.2.self_attn.o_proj.weight`	Linear	1024 × 2048	bfloat16	2,097,152
`model.layers.2.self_attn.q_norm.weight`	Qwen3RMSNorm	128	bfloat16	128

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Layer path	Type	Shape	Dtype	Params
model.layers.2.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.2.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.3	Qwen3DecoderLayer	–	–	–
model.layers.3.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.3.mlp	Qwen3MLP	–	–	–
model.layers.3.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.3.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.3.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.3.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.3.self_attn	Qwen3Attention	–	–	–
model.layers.3.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.3.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.3.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.3.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.3.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.3.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.4	Qwen3DecoderLayer	–	–	–
model.layers.4.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.4.mlp	Qwen3MLP	–	–	–
model.layers.4.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.4.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.4.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.4.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.4.self_attn	Qwen3Attention	–	–	–
model.layers.4.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.4.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.4.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.4.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.4.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.4.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.5	Qwen3DecoderLayer	–	–	–
model.layers.5.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.5.mlp	Qwen3MLP	–	–	–
model.layers.5.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.5.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.5.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.5.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.5.self_attn	Qwen3Attention	–	–	–
model.layers.5.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.5.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.5.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.5.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.5.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.5.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.6	Qwen3DecoderLayer	–	–	–
model.layers.6.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.6.mlp	Qwen3MLP	–	–	–
model.layers.6.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.6.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.6.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.6.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.6.self_attn	Qwen3Attention	–	–	–
model.layers.6.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.6.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.6.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.6.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.6.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.6.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.7	Qwen3DecoderLayer	–	–	–
model.layers.7.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.7.mlp	Qwen3MLP	–	–	–
model.layers.7.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.7.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.7.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.7.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.7.self_attn	Qwen3Attention	–	–	–
model.layers.7.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.7.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.7.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152

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Layer path	Type	Shape	Dtype	Params
model.layers.7.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.7.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.7.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.8	Qwen3DecoderLayer	–	–	–
model.layers.8.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.8.mlp	Qwen3MLP	–	–	–
model.layers.8.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.8.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.8.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.8.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.8.self_attn	Qwen3Attention	–	–	–
model.layers.8.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.8.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.8.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.8.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.8.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.8.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.9	Qwen3DecoderLayer	–	–	–
model.layers.9.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.9.mlp	Qwen3MLP	–	–	–
model.layers.9.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.9.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.9.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.9.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.9.self_attn	Qwen3Attention	–	–	–
model.layers.9.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.9.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.9.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.9.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.9.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.9.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.10	Qwen3DecoderLayer	–	–	–
model.layers.10.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.10.mlp	Qwen3MLP	–	–	–
model.layers.10.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.10.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.10.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.10.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.10.self_attn	Qwen3Attention	–	–	–
model.layers.10.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.10.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.10.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.10.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.10.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.10.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.11	Qwen3DecoderLayer	–	–	–
model.layers.11.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.11.mlp	Qwen3MLP	–	–	–
model.layers.11.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.11.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.11.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.11.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.11.self_attn	Qwen3Attention	–	–	–
model.layers.11.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.11.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.11.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.11.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.11.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.11.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.12	Qwen3DecoderLayer	–	–	–
model.layers.12.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.12.mlp	Qwen3MLP	–	–	–
model.layers.12.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.12.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.12.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.12.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.12.self_attn	Qwen3Attention	–	–	–
model.layers.12.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.12.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576

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Layer path	Type	Shape	Dtype	Params
model.layers.12.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.12.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.12.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.12.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.13	Qwen3DecoderLayer	–	–	–
model.layers.13.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.13.mlp	Qwen3MLP	–	–	–
model.layers.13.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.13.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.13.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.13.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.13.self_attn	Qwen3Attention	–	–	–
model.layers.13.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.13.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.13.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.13.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.13.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.13.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.14	Qwen3DecoderLayer	–	–	–
model.layers.14.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.14.mlp	Qwen3MLP	–	–	–
model.layers.14.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.14.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.14.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.14.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.14.self_attn	Qwen3Attention	–	–	–
model.layers.14.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.14.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.14.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.14.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.14.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.14.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.15	Qwen3DecoderLayer	–	–	–
model.layers.15.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.15.mlp	Qwen3MLP	–	–	–
model.layers.15.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.15.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.15.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.15.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.15.self_attn	Qwen3Attention	–	–	–
model.layers.15.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.15.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.15.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.15.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.15.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.15.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.16	Qwen3DecoderLayer	–	–	–
model.layers.16.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.16.mlp	Qwen3MLP	–	–	–
model.layers.16.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.16.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.16.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.16.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.16.self_attn	Qwen3Attention	–	–	–
model.layers.16.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.16.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.16.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.16.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.16.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.16.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.17	Qwen3DecoderLayer	–	–	–
model.layers.17.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.17.mlp	Qwen3MLP	–	–	–
model.layers.17.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.17.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.17.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.17.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.17.self_attn	Qwen3Attention	–	–	–
model.layers.17.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128

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Layer path	Type	Shape	Dtype	Params
model.layers.17.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.17.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.17.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.17.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.17.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.18	Qwen3DecoderLayer	–	–	–
model.layers.18.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.18.mlp	Qwen3MLP	–	–	–
model.layers.18.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.18.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.18.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.18.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.18.self_attn	Qwen3Attention	–	–	–
model.layers.18.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.18.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.18.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.18.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.18.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.18.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.19	Qwen3DecoderLayer	–	–	–
model.layers.19.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.19.mlp	Qwen3MLP	–	–	–
model.layers.19.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.19.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.19.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.19.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.19.self_attn	Qwen3Attention	–	–	–
model.layers.19.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.19.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.19.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.19.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.19.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.19.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.20	Qwen3DecoderLayer	–	–	–
model.layers.20.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.20.mlp	Qwen3MLP	–	–	–
model.layers.20.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.20.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.20.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.20.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.20.self_attn	Qwen3Attention	–	–	–
model.layers.20.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.20.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.20.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.20.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.20.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.20.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.21	Qwen3DecoderLayer	–	–	–
model.layers.21.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.21.mlp	Qwen3MLP	–	–	–
model.layers.21.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.21.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.21.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.21.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.21.self_attn	Qwen3Attention	–	–	–
model.layers.21.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.21.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.21.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.21.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.21.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.21.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.22	Qwen3DecoderLayer	–	–	–
model.layers.22.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.22.mlp	Qwen3MLP	–	–	–
model.layers.22.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.22.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.22.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.22.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.22.self_attn	Qwen3Attention	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.layers.22.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.22.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.22.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.22.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.22.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.22.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.23	Qwen3DecoderLayer	–	–	–
model.layers.23.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.23.mlp	Qwen3MLP	–	–	–
model.layers.23.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.23.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.23.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.23.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.23.self_attn	Qwen3Attention	–	–	–
model.layers.23.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.23.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.23.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.23.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.23.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.23.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.24	Qwen3DecoderLayer	–	–	–
model.layers.24.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.24.mlp	Qwen3MLP	–	–	–
model.layers.24.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.24.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.24.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.24.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.24.self_attn	Qwen3Attention	–	–	–
model.layers.24.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.24.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.24.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.24.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.24.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.24.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.25	Qwen3DecoderLayer	–	–	–
model.layers.25.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.25.mlp	Qwen3MLP	–	–	–
model.layers.25.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.25.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.25.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.25.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.25.self_attn	Qwen3Attention	–	–	–
model.layers.25.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.25.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.25.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.25.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.25.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.25.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.26	Qwen3DecoderLayer	–	–	–
model.layers.26.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.26.mlp	Qwen3MLP	–	–	–
model.layers.26.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.26.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.26.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.26.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.26.self_attn	Qwen3Attention	–	–	–
model.layers.26.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.26.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.26.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.26.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.26.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.26.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.27	Qwen3DecoderLayer	–	–	–
model.layers.27.input_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024
model.layers.27.mlp	Qwen3MLP	–	–	–
model.layers.27.mlp.down_proj.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.layers.27.mlp.gate_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.27.mlp.up_proj.weight	Linear	3072 × 1024	bfloat16	3,145,728
model.layers.27.post_attention_layernorm.weight	Qwen3RMSNorm	1024	bfloat16	1,024

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Layer path	Type	Shape	Dtype	Params
model.layers.27.self_attn	Qwen3Attention	–	–	–
model.layers.27.self_attn.k_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.27.self_attn.k_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.layers.27.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.layers.27.self_attn.q_norm.weight	Qwen3RMSNorm	128	bfloat16	128
model.layers.27.self_attn.q_proj.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.layers.27.self_attn.v_proj.weight	Linear	1024 × 1024	bfloat16	1,048,576
model.norm.weight	Qwen3RMSNorm	1024	bfloat16	1,024

Appendix B Per-Layer Parameter Count

Module	Params	Running total
model.embed_tokens	155,582,464	155,582,464
model.layers.0.input_layernorm	1,024	155,583,488
model.layers.0.mlp.down_proj	3,145,728	158,729,216
model.layers.0.mlp.gate_proj	3,145,728	161,874,944
model.layers.0.mlp.up_proj	3,145,728	165,020,672
model.layers.0.post_attention_layernorm	1,024	165,021,696
model.layers.0.self_attn.k_norm	128	165,021,824
model.layers.0.self_attn.k_proj	1,048,576	166,070,400
model.layers.0.self_attn.o_proj	2,097,152	168,167,552
model.layers.0.self_attn.q_norm	128	168,167,680
model.layers.0.self_attn.q_proj	2,097,152	170,264,832
model.layers.0.self_attn.v_proj	1,048,576	171,313,408
model.layers.1.input_layernorm	1,024	171,314,432
model.layers.1.mlp.down_proj	3,145,728	174,460,160
model.layers.1.mlp.gate_proj	3,145,728	177,605,888
model.layers.1.mlp.up_proj	3,145,728	180,751,616
model.layers.1.post_attention_layernorm	1,024	180,752,640
model.layers.1.self_attn.k_norm	128	180,752,768
model.layers.1.self_attn.k_proj	1,048,576	181,801,344
model.layers.1.self_attn.o_proj	2,097,152	183,898,496
model.layers.1.self_attn.q_norm	128	183,898,624
model.layers.1.self_attn.q_proj	2,097,152	185,995,776
model.layers.1.self_attn.v_proj	1,048,576	187,044,352
model.layers.2.input_layernorm	1,024	187,045,376
model.layers.2.mlp.down_proj	3,145,728	190,191,104
model.layers.2.mlp.gate_proj	3,145,728	193,336,832
model.layers.2.mlp.up_proj	3,145,728	196,482,560
model.layers.2.post_attention_layernorm	1,024	196,483,584
model.layers.2.self_attn.k_norm	128	196,483,712
model.layers.2.self_attn.k_proj	1,048,576	197,532,288
model.layers.2.self_attn.o_proj	2,097,152	199,629,440
model.layers.2.self_attn.q_norm	128	199,629,568
model.layers.2.self_attn.q_proj	2,097,152	201,726,720
model.layers.2.self_attn.v_proj	1,048,576	202,775,296
model.layers.3.input_layernorm	1,024	202,776,320
model.layers.3.mlp.down_proj	3,145,728	205,922,048
model.layers.3.mlp.gate_proj	3,145,728	209,067,776
model.layers.3.mlp.up_proj	3,145,728	212,213,504
model.layers.3.post_attention_layernorm	1,024	212,214,528
model.layers.3.self_attn.k_norm	128	212,214,656
model.layers.3.self_attn.k_proj	1,048,576	213,263,232
model.layers.3.self_attn.o_proj	2,097,152	215,360,384
model.layers.3.self_attn.q_norm	128	215,360,512
model.layers.3.self_attn.q_proj	2,097,152	217,457,664
model.layers.3.self_attn.v_proj	1,048,576	218,506,240
model.layers.4.input_layernorm	1,024	218,507,264

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Module	Params	Running total
model.layers.4.mlp.down_proj	3,145,728	221,652,992
model.layers.4.mlp.gate_proj	3,145,728	224,798,720
model.layers.4.mlp.up_proj	3,145,728	227,944,448
model.layers.4.post_attention_layernorm	1,024	227,945,472
model.layers.4.self_attn.k_norm	128	227,945,600
model.layers.4.self_attn.k_proj	1,048,576	228,994,176
model.layers.4.self_attn.o_proj	2,097,152	231,091,328
model.layers.4.self_attn.q_norm	128	231,091,456
model.layers.4.self_attn.q_proj	2,097,152	233,188,608
model.layers.4.self_attn.v_proj	1,048,576	234,237,184
model.layers.5.input_layernorm	1,024	234,238,208
model.layers.5.mlp.down_proj	3,145,728	237,383,936
model.layers.5.mlp.gate_proj	3,145,728	240,529,664
model.layers.5.mlp.up_proj	3,145,728	243,675,392
model.layers.5.post_attention_layernorm	1,024	243,676,416
model.layers.5.self_attn.k_norm	128	243,676,544
model.layers.5.self_attn.k_proj	1,048,576	244,725,120
model.layers.5.self_attn.o_proj	2,097,152	246,822,272
model.layers.5.self_attn.q_norm	128	246,822,400
model.layers.5.self_attn.q_proj	2,097,152	248,919,552
model.layers.5.self_attn.v_proj	1,048,576	249,968,128
model.layers.6.input_layernorm	1,024	249,969,152
model.layers.6.mlp.down_proj	3,145,728	253,114,880
model.layers.6.mlp.gate_proj	3,145,728	256,260,608
model.layers.6.mlp.up_proj	3,145,728	259,406,336
model.layers.6.post_attention_layernorm	1,024	259,407,360
model.layers.6.self_attn.k_norm	128	259,407,488
model.layers.6.self_attn.k_proj	1,048,576	260,456,064
model.layers.6.self_attn.o_proj	2,097,152	262,553,216
model.layers.6.self_attn.q_norm	128	262,553,344
model.layers.6.self_attn.q_proj	2,097,152	264,650,496
model.layers.6.self_attn.v_proj	1,048,576	265,699,072
model.layers.7.input_layernorm	1,024	265,700,096
model.layers.7.mlp.down_proj	3,145,728	268,845,824
model.layers.7.mlp.gate_proj	3,145,728	271,991,552
model.layers.7.mlp.up_proj	3,145,728	275,137,280
model.layers.7.post_attention_layernorm	1,024	275,138,304
model.layers.7.self_attn.k_norm	128	275,138,432
model.layers.7.self_attn.k_proj	1,048,576	276,187,008
model.layers.7.self_attn.o_proj	2,097,152	278,284,160
model.layers.7.self_attn.q_norm	128	278,284,288
model.layers.7.self_attn.q_proj	2,097,152	280,381,440
model.layers.7.self_attn.v_proj	1,048,576	281,430,016
model.layers.8.input_layernorm	1,024	281,431,040
model.layers.8.mlp.down_proj	3,145,728	284,576,768
model.layers.8.mlp.gate_proj	3,145,728	287,722,496
model.layers.8.mlp.up_proj	3,145,728	290,868,224
model.layers.8.post_attention_layernorm	1,024	290,869,248
model.layers.8.self_attn.k_norm	128	290,869,376
model.layers.8.self_attn.k_proj	1,048,576	291,917,952
model.layers.8.self_attn.o_proj	2,097,152	294,015,104
model.layers.8.self_attn.q_norm	128	294,015,232
model.layers.8.self_attn.q_proj	2,097,152	296,112,384
model.layers.8.self_attn.v_proj	1,048,576	297,160,960
model.layers.9.input_layernorm	1,024	297,161,984
model.layers.9.mlp.down_proj	3,145,728	300,307,712
model.layers.9.mlp.gate_proj	3,145,728	303,453,440
model.layers.9.mlp.up_proj	3,145,728	306,599,168

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Module	Params	Running total
model.layers.9.post_attention_layernorm	1,024	306,600,192
model.layers.9.self_attn.k_norm	128	306,600,320
model.layers.9.self_attn.k_proj	1,048,576	307,648,896
model.layers.9.self_attn.o_proj	2,097,152	309,746,048
model.layers.9.self_attn.q_norm	128	309,746,176
model.layers.9.self_attn.q_proj	2,097,152	311,843,328
model.layers.9.self_attn.v_proj	1,048,576	312,891,904
model.layers.10.input_layernorm	1,024	312,892,928
model.layers.10.mlp.down_proj	3,145,728	316,038,656
model.layers.10.mlp.gate_proj	3,145,728	319,184,384
model.layers.10.mlp.up_proj	3,145,728	322,330,112
model.layers.10.post_attention_layernorm	1,024	322,331,136
model.layers.10.self_attn.k_norm	128	322,331,264
model.layers.10.self_attn.k_proj	1,048,576	323,379,840
model.layers.10.self_attn.o_proj	2,097,152	325,476,992
model.layers.10.self_attn.q_norm	128	325,477,120
model.layers.10.self_attn.q_proj	2,097,152	327,574,272
model.layers.10.self_attn.v_proj	1,048,576	328,622,848
model.layers.11.input_layernorm	1,024	328,623,872
model.layers.11.mlp.down_proj	3,145,728	331,769,600
model.layers.11.mlp.gate_proj	3,145,728	334,915,328
model.layers.11.mlp.up_proj	3,145,728	338,061,056
model.layers.11.post_attention_layernorm	1,024	338,062,080
model.layers.11.self_attn.k_norm	128	338,062,208
model.layers.11.self_attn.k_proj	1,048,576	339,110,784
model.layers.11.self_attn.o_proj	2,097,152	341,207,936
model.layers.11.self_attn.q_norm	128	341,208,064
model.layers.11.self_attn.q_proj	2,097,152	343,305,216
model.layers.11.self_attn.v_proj	1,048,576	344,353,792
model.layers.12.input_layernorm	1,024	344,354,816
model.layers.12.mlp.down_proj	3,145,728	347,500,544
model.layers.12.mlp.gate_proj	3,145,728	350,646,272
model.layers.12.mlp.up_proj	3,145,728	353,792,000
model.layers.12.post_attention_layernorm	1,024	353,793,024
model.layers.12.self_attn.k_norm	128	353,793,152
model.layers.12.self_attn.k_proj	1,048,576	354,841,728
model.layers.12.self_attn.o_proj	2,097,152	356,938,880
model.layers.12.self_attn.q_norm	128	356,939,008
model.layers.12.self_attn.q_proj	2,097,152	359,036,160
model.layers.12.self_attn.v_proj	1,048,576	360,084,736
model.layers.13.input_layernorm	1,024	360,085,760
model.layers.13.mlp.down_proj	3,145,728	363,231,488
model.layers.13.mlp.gate_proj	3,145,728	366,377,216
model.layers.13.mlp.up_proj	3,145,728	369,522,944
model.layers.13.post_attention_layernorm	1,024	369,523,968
model.layers.13.self_attn.k_norm	128	369,524,096
model.layers.13.self_attn.k_proj	1,048,576	370,572,672
model.layers.13.self_attn.o_proj	2,097,152	372,669,824
model.layers.13.self_attn.q_norm	128	372,669,952
model.layers.13.self_attn.q_proj	2,097,152	374,767,104
model.layers.13.self_attn.v_proj	1,048,576	375,815,680
model.layers.14.input_layernorm	1,024	375,816,704
model.layers.14.mlp.down_proj	3,145,728	378,962,432
model.layers.14.mlp.gate_proj	3,145,728	382,108,160
model.layers.14.mlp.up_proj	3,145,728	385,253,888
model.layers.14.post_attention_layernorm	1,024	385,254,912
model.layers.14.self_attn.k_norm	128	385,255,040
model.layers.14.self_attn.k_proj	1,048,576	386,303,616

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Module	Params	Running total
model.layers.14.self_attn.o_proj	2,097,152	388,400,768
model.layers.14.self_attn.q_norm	128	388,400,896
model.layers.14.self_attn.q_proj	2,097,152	390,498,048
model.layers.14.self_attn.v_proj	1,048,576	391,546,624
model.layers.15.input_layernorm	1,024	391,547,648
model.layers.15.mlp.down_proj	3,145,728	394,693,376
model.layers.15.mlp.gate_proj	3,145,728	397,839,104
model.layers.15.mlp.up_proj	3,145,728	400,984,832
model.layers.15.post_attention_layernorm	1,024	400,985,856
model.layers.15.self_attn.k_norm	128	400,985,984
model.layers.15.self_attn.k_proj	1,048,576	402,034,560
model.layers.15.self_attn.o_proj	2,097,152	404,131,712
model.layers.15.self_attn.q_norm	128	404,131,840
model.layers.15.self_attn.q_proj	2,097,152	406,228,992
model.layers.15.self_attn.v_proj	1,048,576	407,277,568
model.layers.16.input_layernorm	1,024	407,278,592
model.layers.16.mlp.down_proj	3,145,728	410,424,320
model.layers.16.mlp.gate_proj	3,145,728	413,570,048
model.layers.16.mlp.up_proj	3,145,728	416,715,776
model.layers.16.post_attention_layernorm	1,024	416,716,800
model.layers.16.self_attn.k_norm	128	416,716,928
model.layers.16.self_attn.k_proj	1,048,576	417,765,504
model.layers.16.self_attn.o_proj	2,097,152	419,862,656
model.layers.16.self_attn.q_norm	128	419,862,784
model.layers.16.self_attn.q_proj	2,097,152	421,959,936
model.layers.16.self_attn.v_proj	1,048,576	423,008,512
model.layers.17.input_layernorm	1,024	423,009,536
model.layers.17.mlp.down_proj	3,145,728	426,155,264
model.layers.17.mlp.gate_proj	3,145,728	429,300,992
model.layers.17.mlp.up_proj	3,145,728	432,446,720
model.layers.17.post_attention_layernorm	1,024	432,447,744
model.layers.17.self_attn.k_norm	128	432,447,872
model.layers.17.self_attn.k_proj	1,048,576	433,496,448
model.layers.17.self_attn.o_proj	2,097,152	435,593,600
model.layers.17.self_attn.q_norm	128	435,593,728
model.layers.17.self_attn.q_proj	2,097,152	437,690,880
model.layers.17.self_attn.v_proj	1,048,576	438,739,456
model.layers.18.input_layernorm	1,024	438,740,480
model.layers.18.mlp.down_proj	3,145,728	441,886,208
model.layers.18.mlp.gate_proj	3,145,728	445,031,936
model.layers.18.mlp.up_proj	3,145,728	448,177,664
model.layers.18.post_attention_layernorm	1,024	448,178,688
model.layers.18.self_attn.k_norm	128	448,178,816
model.layers.18.self_attn.k_proj	1,048,576	449,227,392
model.layers.18.self_attn.o_proj	2,097,152	451,324,544
model.layers.18.self_attn.q_norm	128	451,324,672
model.layers.18.self_attn.q_proj	2,097,152	453,421,824
model.layers.18.self_attn.v_proj	1,048,576	454,470,400
model.layers.19.input_layernorm	1,024	454,471,424
model.layers.19.mlp.down_proj	3,145,728	457,617,152
model.layers.19.mlp.gate_proj	3,145,728	460,762,880
model.layers.19.mlp.up_proj	3,145,728	463,908,608
model.layers.19.post_attention_layernorm	1,024	463,909,632
model.layers.19.self_attn.k_norm	128	463,909,760
model.layers.19.self_attn.k_proj	1,048,576	464,958,336
model.layers.19.self_attn.o_proj	2,097,152	467,055,488
model.layers.19.self_attn.q_norm	128	467,055,616
model.layers.19.self_attn.q_proj	2,097,152	469,152,768

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Module	Params	Running total
model.layers.19.self_attn.v_proj	1,048,576	470,201,344
model.layers.20.input_layernorm	1,024	470,202,368
model.layers.20.mlp.down_proj	3,145,728	473,348,096
model.layers.20.mlp.gate_proj	3,145,728	476,493,824
model.layers.20.mlp.up_proj	3,145,728	479,639,552
model.layers.20.post_attention_layernorm	1,024	479,640,576
model.layers.20.self_attn.k_norm	128	479,640,704
model.layers.20.self_attn.k_proj	1,048,576	480,689,280
model.layers.20.self_attn.o_proj	2,097,152	482,786,432
model.layers.20.self_attn.q_norm	128	482,786,560
model.layers.20.self_attn.q_proj	2,097,152	484,883,712
model.layers.20.self_attn.v_proj	1,048,576	485,932,288
model.layers.21.input_layernorm	1,024	485,933,312
model.layers.21.mlp.down_proj	3,145,728	489,079,040
model.layers.21.mlp.gate_proj	3,145,728	492,224,768
model.layers.21.mlp.up_proj	3,145,728	495,370,496
model.layers.21.post_attention_layernorm	1,024	495,371,520
model.layers.21.self_attn.k_norm	128	495,371,648
model.layers.21.self_attn.k_proj	1,048,576	496,420,224
model.layers.21.self_attn.o_proj	2,097,152	498,517,376
model.layers.21.self_attn.q_norm	128	498,517,504
model.layers.21.self_attn.q_proj	2,097,152	500,614,656
model.layers.21.self_attn.v_proj	1,048,576	501,663,232
model.layers.22.input_layernorm	1,024	501,664,256
model.layers.22.mlp.down_proj	3,145,728	504,809,984
model.layers.22.mlp.gate_proj	3,145,728	507,955,712
model.layers.22.mlp.up_proj	3,145,728	511,101,440
model.layers.22.post_attention_layernorm	1,024	511,102,464
model.layers.22.self_attn.k_norm	128	511,102,592
model.layers.22.self_attn.k_proj	1,048,576	512,151,168
model.layers.22.self_attn.o_proj	2,097,152	514,248,320
model.layers.22.self_attn.q_norm	128	514,248,448
model.layers.22.self_attn.q_proj	2,097,152	516,345,600
model.layers.22.self_attn.v_proj	1,048,576	517,394,176
model.layers.23.input_layernorm	1,024	517,395,200
model.layers.23.mlp.down_proj	3,145,728	520,540,928
model.layers.23.mlp.gate_proj	3,145,728	523,686,656
model.layers.23.mlp.up_proj	3,145,728	526,832,384
model.layers.23.post_attention_layernorm	1,024	526,833,408
model.layers.23.self_attn.k_norm	128	526,833,536
model.layers.23.self_attn.k_proj	1,048,576	527,882,112
model.layers.23.self_attn.o_proj	2,097,152	529,979,264
model.layers.23.self_attn.q_norm	128	529,979,392
model.layers.23.self_attn.q_proj	2,097,152	532,076,544
model.layers.23.self_attn.v_proj	1,048,576	533,125,120
model.layers.24.input_layernorm	1,024	533,126,144
model.layers.24.mlp.down_proj	3,145,728	536,271,872
model.layers.24.mlp.gate_proj	3,145,728	539,417,600
model.layers.24.mlp.up_proj	3,145,728	542,563,328
model.layers.24.post_attention_layernorm	1,024	542,564,352
model.layers.24.self_attn.k_norm	128	542,564,480
model.layers.24.self_attn.k_proj	1,048,576	543,613,056
model.layers.24.self_attn.o_proj	2,097,152	545,710,208
model.layers.24.self_attn.q_norm	128	545,710,336
model.layers.24.self_attn.q_proj	2,097,152	547,807,488
model.layers.24.self_attn.v_proj	1,048,576	548,856,064
model.layers.25.input_layernorm	1,024	548,857,088
model.layers.25.mlp.down_proj	3,145,728	552,002,816

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Module	Params	Running total
model.layers.25.mlp.gate_proj	3,145,728	555,148,544
model.layers.25.mlp.up_proj	3,145,728	558,294,272
model.layers.25.post_attention_layernorm	1,024	558,295,296
model.layers.25.self_attn.k_norm	128	558,295,424
model.layers.25.self_attn.k_proj	1,048,576	559,344,000
model.layers.25.self_attn.o_proj	2,097,152	561,441,152
model.layers.25.self_attn.q_norm	128	561,441,280
model.layers.25.self_attn.q_proj	2,097,152	563,538,432
model.layers.25.self_attn.v_proj	1,048,576	564,587,008
model.layers.26.input_layernorm	1,024	564,588,032
model.layers.26.mlp.down_proj	3,145,728	567,733,760
model.layers.26.mlp.gate_proj	3,145,728	570,879,488
model.layers.26.mlp.up_proj	3,145,728	574,025,216
model.layers.26.post_attention_layernorm	1,024	574,026,240
model.layers.26.self_attn.k_norm	128	574,026,368
model.layers.26.self_attn.k_proj	1,048,576	575,074,944
model.layers.26.self_attn.o_proj	2,097,152	577,172,096
model.layers.26.self_attn.q_norm	128	577,172,224
model.layers.26.self_attn.q_proj	2,097,152	579,269,376
model.layers.26.self_attn.v_proj	1,048,576	580,317,952
model.layers.27.input_layernorm	1,024	580,318,976
model.layers.27.mlp.down_proj	3,145,728	583,464,704
model.layers.27.mlp.gate_proj	3,145,728	586,610,432
model.layers.27.mlp.up_proj	3,145,728	589,756,160
model.layers.27.post_attention_layernorm	1,024	589,757,184
model.layers.27.self_attn.k_norm	128	589,757,312
model.layers.27.self_attn.k_proj	1,048,576	590,805,888
model.layers.27.self_attn.o_proj	2,097,152	592,903,040
model.layers.27.self_attn.q_norm	128	592,903,168
model.layers.27.self_attn.q_proj	2,097,152	595,000,320
model.layers.27.self_attn.v_proj	1,048,576	596,048,896
model.norm	1,024	596,049,920